

Case Study #1: Sales Forecasting

Solutions to the Case

In this case, consider the data on worldwide monthly sales from a large chain of grocery stores (673_case1.csv). The monthly sales are given for the period of 2016-2025 and measured in millions of dollars. The goal is to identify the best forecasting model to predict monthly sales in 12 months of 2026.

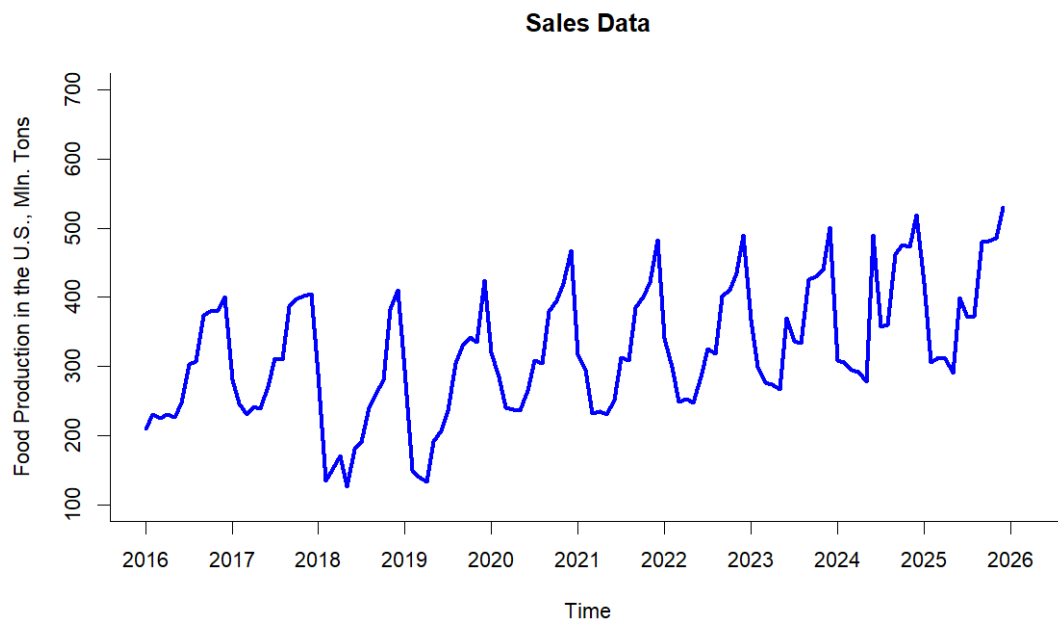
Questions

1. Identify time series components and plot the data.

1a. Create time series data set in R using the `ts()` function.

(Explanation is optional and can vary). We created a time series data set `sales.ts` from the data frame `sales.data` using the `ts()` function from January 2016 (month 1) through December of 2025 (month 12) with a frequency of 12 for monthly data.

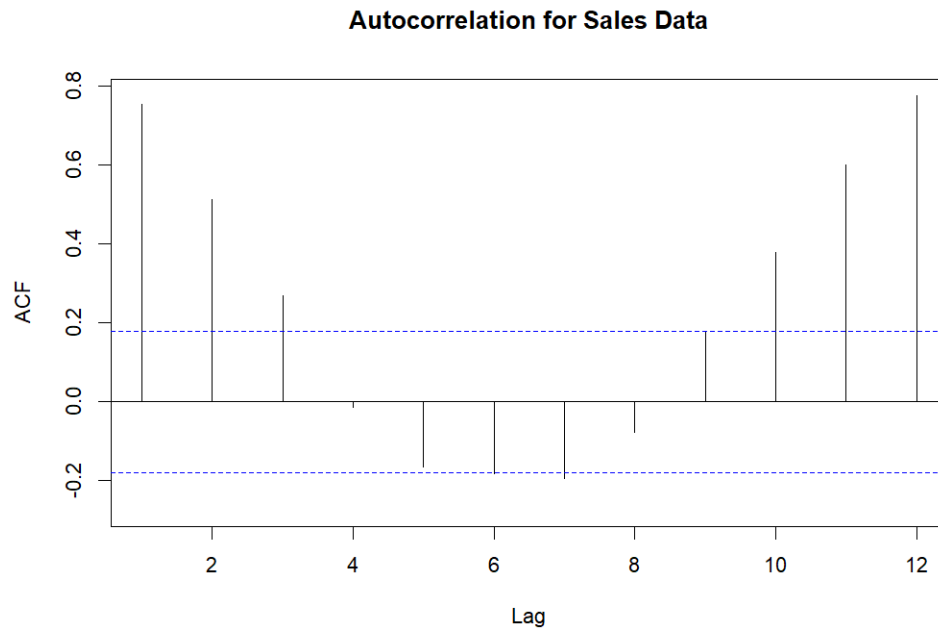
1b. Employ the `plot()` function to create a data plot with the historical data, provide it in your report, and explain what data patterns can be visualized in this plot.



The data plot above shows a potential trend and additive seasonality with the lower sales at the beginning of each year (approximately January-March) and peak sales at the end of each year (approximately October-December).

1c. Apply the `Acf()` function to identify possible time series components. Provide in the report the autocorrelation chart and explain the time series components existing in the historical data.

The autocorrelation chart is shown below:



A positive autocorrelation coefficient in lag 1 is substantially higher than the horizontal threshold (significantly greater than zero) and in several lags gradually drops to zero, which is indicative of an upward trend component and also of the level component in the sales dataset. A positive autocorrelation coefficient in lag 12, which is also statistically significant (greater than zero), points to monthly seasonality.

2. Use trailing MA for forecasting time series.

2a. Develop data partition with the validation partition of 36 monthly periods (3 years) and training partition of 84 monthly periods (7 years). Provide the data partition's R code in your report.

The code to develop the data partition is shown below.

```
nValid <- 36
nTrain <- length(prod.ts) - nValid
train.ts <- window(prod.ts, start = c(2015, 1), end = c(2015, nTrain))
valid.ts <- window(prod.ts, start = c(2015, nTrain + 1),
                  end = c(2015, nTrain + nValid))
```

2b. Use the *rollmean()* function to develop 3 trailing MAs with the window width of 3, 5, and 12 for the training partition. Present the R code for these MAs in your report.

```
# Create trailing moving average with window widths of k = 3, 5, and 12.
# In rollmean(), use argument align = "right" to calculate a trailing MA.
ma.trailing_3 <- rollmean(train.ts, k = 3, align = "right")
ma.trailing_5 <- rollmean(train.ts, k = 5, align = "right")
ma.trailing_12 <- rollmean(train.ts, k = 12, align = "right")
```

2c. Use the *forecast()* function to create a trailing MA forecast for each window width from question 2b in the validation period, and present one of them, e.g., with window width of 3, in your report.

The trailing MA forecast with window width of 3 in the validation period is presented below (with no confidence interval).

	Point	Forecast	Lo 0	Hi 0
Jan 2023		422.7012	422.7012	422.7012
Feb 2023		370.3600	370.3600	370.3600
Mar 2023		289.5857	289.5857	289.5857
Apr 2023		264.3034	264.3034	264.3034
May 2023		255.4876	255.4876	255.4876
Jun 2023		266.3138	266.3138	266.3138
Jul 2023		288.7483	288.7483	288.7483
Aug 2023		317.2277	317.2277	317.2277
Sep 2023		356.3758	356.3758	356.3758
Oct 2023		385.7681	385.7681	385.7681
Nov 2023		419.5818	419.5818	419.5818
Dec 2023		445.4997	445.4997	445.4997
Jan 2024		422.7012	422.7012	422.7012
Feb 2024		370.3600	370.3600	370.3600
Mar 2024		289.5857	289.5857	289.5857
Apr 2024		264.3034	264.3034	264.3034
May 2024		255.4876	255.4876	255.4876
Jun 2024		266.3138	266.3138	266.3138
Jul 2024		288.7483	288.7483	288.7483
Aug 2024		317.2277	317.2277	317.2277
Sep 2024		356.3758	356.3758	356.3758
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Dec 2024		445.4997	445.4997	445.4997
Jan 2025		422.7012	422.7012	422.7012
Feb 2025		370.3600	370.3600	370.3600
Mar 2025		289.5857	289.5857	289.5857
Apr 2025		264.3034	264.3034	264.3034
May 2025		255.4876	255.4876	255.4876
Jun 2025		266.3138	266.3138	266.3138
Jul 2025		288.7483	288.7483	288.7483
Aug 2025		317.2277	317.2277	317.2277
Sep 2025		356.3758	356.3758	356.3758
Oct 2025		385.7681	385.7681	385.7681
Nov 2025		419.5818	419.5818	419.5818
Dec 2025		445.4997	445.4997	445.4997

2d. Apply the *accuracy()* function to compare accuracy of the 3 trailing MA forecasts in the validation period. Present the accuracy measures in your report, compare MAPE and RMSE of these forecasts, and identify the best trailing MA forecast.

Trailing MA with windows width of 3:

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	40.251	75.668	61.688	8.787	15.655	0.398	1.073

Trailing MA with windows width of 5:

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	43.83	94.615	82.081	8.109	20.829	0.599	1.29

Trailing MA with windows width of 12:

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	27.706	83.213	69.739	3.149	17.61	0.528	1.047

Comparing the MAPE and RMSE of the 3 forecasts for the validation period – trailing MA with windows width of 3, 5, and 12 – we can see that the most accurate forecast here is the trailing MA forecast with window width of 3, which has the lowest MAPE of 15.66% and the lowest RMSE of 75.69.

3. Apply the two-level forecast with regression and trailing MA for residuals.

3a. Develop using the *tslm()* function a regression model with linear trend and seasonality. Present the model summary in your report. Present and briefly explain the model equation in your report. Using this model, forecast food production in the validation period with the *forecast()* function. Present the forecast in your report.

The summary of the developed regression model in the training period (partition) is presented below.

```
Call:
tslm(formula = train.ts ~ trend + season)

Residuals:
    Min       1Q   Median       3Q      Max
-91.03 -15.90  11.34   24.02  45.42

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  265.8600    16.8619   15.767 < 2e-16 ***
trend         0.7347     0.1853    3.965 0.000173 ***
season2     -59.6061    21.7870   -2.736 0.007854 **
season3     -84.9122    21.7894   -3.897 0.000218 ***
season4     -81.1326    21.7933   -3.723 0.000391 ***
season5     -82.2815    21.7988   -3.775 0.000330 ***
season6     -53.0019    21.8059   -2.431 0.017601 *
season7     -13.8223    21.8146   -0.634 0.528360
season8       0.7573    21.8248    0.035 0.972417
season9      60.6226    21.8366    2.776 0.007027 **
season10     72.5022    21.8500    3.318 0.001431 **
season11     96.2676    21.8649    4.403 3.70e-05 ***
season12    138.8615    21.8814    6.346 1.81e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.76 on 71 degrees of freedom
Multiple R-squared:  0.8106, Adjusted R-squared:  0.7785
F-statistic: 25.32 on 12 and 71 DF, p-value: < 2.2e-16
```

The regression model contains 12 predictors including the trend and 11 seasonal dummy variables for February (sason2) through December (season12). The regression equation (coefficients are rounded to 2 decimals) is:

$$y_t = 265.86 - 0.73 t - 59.61 D_2 - 84.91 D_3 + \dots + 138.86 D_{12}$$

The production forecast using this regression model in the validation period is presented below (confidence interval is not used):

	Point	Forecast	Lo 0	Hi 0
Jan 2023		328.3071	328.3071	328.3071
Feb 2023		269.4357	269.4357	269.4357
Mar 2023		244.8643	244.8643	244.8643
Apr 2023		249.3786	249.3786	249.3786
May 2023		248.9643	248.9643	248.9643
Jun 2023		278.9786	278.9786	278.9786
Jul 2023		318.8929	318.8929	318.8929
Aug 2023		334.2071	334.2071	334.2071
Sep 2023		394.8071	394.8071	394.8071
Oct 2023		407.4214	407.4214	407.4214
Nov 2023		431.9214	431.9214	431.9214
Dec 2023		475.2500	475.2500	475.2500
Jan 2024		337.1232	337.1232	337.1232
Feb 2024		278.2518	278.2518	278.2518
Mar 2024		253.6804	253.6804	253.6804
Apr 2024		258.1946	258.1946	258.1946
May 2024		257.7804	257.7804	257.7804
Jun 2024		287.7946	287.7946	287.7946
Jul 2024		327.7089	327.7089	327.7089
Aug 2024		343.0232	343.0232	343.0232
Sep 2024		403.6232	403.6232	403.6232
Oct 2024		416.2375	416.2375	416.2375
Nov 2024		440.7375	440.7375	440.7375

Dec 2024	484.0661	484.0661	484.0661
Jan 2025	345.9393	345.9393	345.9393
Feb 2025	287.0679	287.0679	287.0679
Mar 2025	262.4964	262.4964	262.4964
Apr 2025	267.0107	267.0107	267.0107
May 2025	266.5964	266.5964	266.5964
Jun 2025	296.6107	296.6107	296.6107
Jul 2025	336.5250	336.5250	336.5250
Aug 2025	351.8393	351.8393	351.8393
Sep 2025	412.4393	412.4393	412.4393
Oct 2025	425.0536	425.0536	425.0536
Nov 2025	449.5536	449.5536	449.5536
Dec 2025	492.8821	492.8821	492.8821

3b. Identify regression residuals in the training period, apply a trailing MA (window width of 3) for these residuals using the *rollmean()* function, and identify trailing MA forecast of these residuals in the validation period (use the *forecast()* function).

The trailing MA forecast (window width of 3) for the regression residuals in the validation period is presented below:

	Point	Forecast	Lo 0	Hi 0
Jan 2023	16.11819	16.11819	16.11819	16.11819
Feb 2023	16.11819	16.11819	16.11819	16.11819
Mar 2023	16.11819	16.11819	16.11819	16.11819
Apr 2023	16.11819	16.11819	16.11819	16.11819
May 2023	16.11819	16.11819	16.11819	16.11819
Jun 2023	16.11819	16.11819	16.11819	16.11819
Jul 2023	16.11819	16.11819	16.11819	16.11819
Aug 2023	16.11819	16.11819	16.11819	16.11819
Sep 2023	16.11819	16.11819	16.11819	16.11819
Oct 2023	16.11819	16.11819	16.11819	16.11819
Nov 2023	16.11819	16.11819	16.11819	16.11819
Dec 2023	16.11819	16.11819	16.11819	16.11819
Jan 2024	16.11819	16.11819	16.11819	16.11819
Feb 2024	16.11819	16.11819	16.11819	16.11819
Mar 2024	16.11819	16.11819	16.11819	16.11819
Apr 2024	16.11819	16.11819	16.11819	16.11819
May 2024	16.11819	16.11819	16.11819	16.11819
Jun 2024	16.11819	16.11819	16.11819	16.11819
Jul 2024	16.11819	16.11819	16.11819	16.11819
Aug 2024	16.11819	16.11819	16.11819	16.11819
Sep 2024	16.11819	16.11819	16.11819	16.11819
Oct 2024	16.11819	16.11819	16.11819	16.11819
Nov 2024	16.11819	16.11819	16.11819	16.11819
Dec 2024	16.11819	16.11819	16.11819	16.11819
Jan 2025	16.11819	16.11819	16.11819	16.11819
Feb 2025	16.11819	16.11819	16.11819	16.11819
Mar 2025	16.11819	16.11819	16.11819	16.11819
Apr 2025	16.11819	16.11819	16.11819	16.11819
May 2025	16.11819	16.11819	16.11819	16.11819
Jun 2025	16.11819	16.11819	16.11819	16.11819
Jul 2025	16.11819	16.11819	16.11819	16.11819
Aug 2025	16.11819	16.11819	16.11819	16.11819
Sep 2025	16.11819	16.11819	16.11819	16.11819
Oct 2025	16.11819	16.11819	16.11819	16.11819
Nov 2025	16.11819	16.11819	16.11819	16.11819
Dec 2025	16.11819	16.11819	16.11819	16.11819

3c. Develop a two-level forecast for the validation period by combining the regression forecast and trailing MA forecast for residuals. Present in your report a table that contains validation data, regression forecast, trailing MA forecast for residuals, and two-level (combined) forecast in the validation period. Apply the *accuracy()* function to compare accuracy of the regression model with linear trend and seasonality and the two-level (combined) model with regression and trailing MA for residuals. Present the accuracy measures in your report, compare MAPE and RMSE of these forecasts, and identify the best forecasting model for the validation period.

The table below contains validation partition data (*Sales*), regression forecast (*Regression.Fst*), MA forecast for regression residuals (*MA.Residuals.Fst*), and the combined (2-level) forecast (*Combined.Fst*) that combines the two previous forecasts:

	Sales	Regression.Fst	MA.Residuals.Fst	Combined.Fst
1	367.6	328.3071	16.11819	344.4253
2	299.0	269.4357	16.11819	285.5539
3	275.8	244.8643	16.11819	260.9825
4	273.0	249.3786	16.11819	265.4968
5	267.2	248.9643	16.11819	265.0825
6	369.4	278.9786	16.11819	295.0968
7	335.8	318.8929	16.11819	335.0110
8	334.0	334.2071	16.11819	350.3253
9	426.1	394.8071	16.11819	410.9253
10	430.3	407.4214	16.11819	423.5396
11	439.6	431.9214	16.11819	448.0396
12	501.4	475.2500	16.11819	491.3682
13	308.7	337.1232	16.11819	353.2414
14	305.2	278.2518	16.11819	294.3700
15	294.4	253.6804	16.11819	269.7985
16	291.5	258.1946	16.11819	274.3128
17	277.5	257.7804	16.11819	273.8985
18	490.2	287.7946	16.11819	303.9128
19	357.9	327.7089	16.11819	343.8271
20	360.7	343.0232	16.11819	359.1414
21	461.7	403.6232	16.11819	419.7414
22	476.2	416.2375	16.11819	432.3557
23	472.8	440.7375	16.11819	456.8557
24	518.3	484.0661	16.11819	500.1843
25	420.3	345.9393	16.11819	362.0575
26	306.3	287.0679	16.11819	303.1860
27	313.2	262.4964	16.11819	278.6146
28	310.9	267.0107	16.11819	283.1289
29	290.3	266.5964	16.11819	282.7146
30	399.7	296.6107	16.11819	312.7289
31	371.1	336.5250	16.11819	352.6432
32	372.7	351.8393	16.11819	367.9575
33	479.2	412.4393	16.11819	428.5575
34	481.5	425.0536	16.11819	441.1718
35	485.8	449.5536	16.11819	465.6718
36	529.6	492.8821	16.11819	509.0003

The accuracy performance measure for the regression and two-level forecasts in the validation period are presented below:

Regression Model with Linear Trend and Seasonality:

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	39.729	53.889	41.319	10.131	10.646	-0.097	0.831

Two-Level Model-Regression Model and Trailing MA for Residuals:

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	23.611	43.395	27.461	5.695	6.875	-0.097	0.691

Out of the two forecasting models in the validation period, the most accurate model is the two-level model, which produced the lowest values for MAPE (6.88%) and RMSE (43.40).

3d. For the entire data set, identify the regression model with linear trend and seasonality and the trailing MA with the window width of 3 for the regression residuals. Use these models to forecast 12 future months of 2026 and develop a two-level forecast for the 12 future months as a combination of the specified forecasts. Present in your report a table that contains regression forecast, trailing MA forecast for residuals, and two-level (combined) forecast in the 12 future months of 2026.

Based on the entire data set, we developed the regression model with linear trend and seasonality. We also identified the regression residuals for the entire data set, and trailing MA residuals (window width of 3) for the entire data set. Then, we develop the regression forecast (*Regression.Fst*) and trailing MA forecast, window width of 6 (*MA.Residuals.Fst*) for the 12 months of 2026. Finally, we developed the two-

level (*Combined.Fst*) forecast as a combination of the two previous forecasts. See all these forecasts in the table below.

	Regression.Fst	MA.Residuals.Fst	Combined.Fst
1	392.3653	8.926791	401.2921
2	332.5453	8.926791	341.4721
3	312.6353	8.926791	321.5621
4	314.9953	8.926791	323.9221
5	310.6653	8.926791	319.5921
6	374.1053	8.926791	383.0321
7	382.5953	8.926791	391.5221
8	393.5753	8.926791	402.5021
9	465.9553	8.926791	474.8821
10	476.8853	8.926791	485.8121
11	495.0553	8.926791	503.9821
12	540.4953	8.926791	549.4221

3e. Develop a seasonal naïve forecast for the entire historical data set and apply the *accuracy()* function to compare accuracy of the three forecasting models: seasonal naïve forecast, regression model with trend and seasonality, and two-level (combined) model with regression and trailing MA for residuals. Present the accuracy measures in your report, compare the MAPE and RMSE of these forecasts, and identify the best forecasting model for forecasting monthly sales in 2026.

The accuracy performance measures for the three forecasts are presented below:

Regression Model with Linear Trend and Seasonality:

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	0	37.441	26.504	-2.257	10.646	0.562	0.89

Two-Level Model – Regression and Trailing MA for Residuals:

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	-0.072	20.787	14.751	-0.674	5.258	-0.086	0.404

Seasonal Naïve Model:

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	11.556	48.445	33.769	1.473	12.793	0.598	1.047

Out of the three forecasting models – regression with trend and seasonality, two-level model (regression with trend and seasonality and trailing MA for residuals), and seasonal naïve forecast – the most accurate model is the two-level model, which produced the lowest values for MAPE (5.26%) and RMSE (20.79).

4. Use advanced exponential smoothing methods.

4a. For the training partition (from question 2a), use the *ets()* function to develop a Holt-Winter's (HW) model with automated selection of error, trend, and seasonality options, and automated selection of smoothing parameters for the training partition. Present and explain the model in your report. Use the model to forecast food production for the validation period using the *forecast()* function, and present this forecast in your report.

A summary of the Holt-Winter's (HW) model with the automated selection of the model options and automated selection of the smoothing parameters for the training period is shown below:

```
ETS(A,N,A)
```

```
Call:
```

```
ets(y = train.ts, model = "ZZZ")
```

```
Smoothing parameters:
```

```
alpha = 0.5146
```

```
gamma = 1e-04
```

Initial states:

```
l = 300.5517
s = 144.7238 98.7915 73.4625 64.6324 1.3655 -14.8298
    -52.7936 -84.167 -82.698 -87.0975 -60.3795 -1.0103
```

sigma: 29.6521

```
      AIC      AICC      BIC
956.3153 963.3741 992.7775
```

This HW model has the (A, N, A) options, i.e., additive error, no trend, and additive seasonality. The optimal value for exponential smoothing constant (alpha) is 0.5146, no smoothing constant for trend estimate (beta = 0), and smoothing constant for seasonality estimate (gamma) is 0.0001. The HW model's forecast in the validation period is presented below:

	Point	Forecast	Lo 0	Hi 0
Jan 2023		339.8050	339.8050	339.8050
Feb 2023		280.4347	280.4347	280.4347
Mar 2023		253.7176	253.7176	253.7176
Apr 2023		258.1159	258.1159	258.1159
May 2023		256.6470	256.6470	256.6470
Jun 2023		288.0185	288.0185	288.0185
Jul 2023		325.9841	325.9841	325.9841
Aug 2023		342.1779	342.1779	342.1779
Sep 2023		405.4426	405.4426	405.4426
Oct 2023		414.2771	414.2771	414.2771
Nov 2023		439.6043	439.6043	439.6043
Dec 2023		485.5343	485.5343	485.5343
Jan 2024		339.8050	339.8050	339.8050
Feb 2024		280.4347	280.4347	280.4347
Mar 2024		253.7176	253.7176	253.7176
Apr 2024		258.1159	258.1159	258.1159
May 2024		256.6470	256.6470	256.6470
Jun 2024		288.0185	288.0185	288.0185
Jul 2024		325.9841	325.9841	325.9841
Aug 2024		342.1779	342.1779	342.1779
Sep 2024		405.4426	405.4426	405.4426
Oct 2024		414.2771	414.2771	414.2771
Nov 2024		439.6043	439.6043	439.6043
Dec 2024		485.5343	485.5343	485.5343
Jan 2025		339.8050	339.8050	339.8050
Feb 2025		280.4347	280.4347	280.4347
Mar 2025		253.7176	253.7176	253.7176
Apr 2025		258.1159	258.1159	258.1159
May 2025		256.6470	256.6470	256.6470
Jun 2025		288.0185	288.0185	288.0185
Jul 2025		325.9841	325.9841	325.9841
Aug 2025		342.1779	342.1779	342.1779
Sep 2025		405.4426	405.4426	405.4426
Oct 2025		414.2771	414.2771	414.2771
Nov 2025		439.6043	439.6043	439.6043
Dec 2025		485.5343	485.5343	485.5343

4b. To make a forecast for the 12 future months of 2026, use the entire data set (no partitioning) to develop the HW model using the *ets()* function for the model with the automated selection of error, trend, and seasonality options, and automated selection of smoothing parameters. Present and explain this model in your report. Use the model to forecast monthly sales for the 12 future months of 2026 using the *forecast()* function, and present the forecast in your report.

Using the entire data set, we received the following Holt-Winter's model with the automated selection of the model options and optimal smoothing parameters:

ETS(A,N,A)

```
Call:
ets(y = sales.ts, model = "zzz")
```

```

Smoothing parameters:
  alpha = 0.3601
  gamma = 1e-04

Initial states:
  l = 301.3873
  s = 136.8607 92.772 76.0252 68.9128 -5.9108 -17.7831
      -22.862 -89.2209 -84.0142 -85.0993 -63.2902 -6.3904

sigma: 32.3405

      AIC      AICC      BIC
1423.929 1428.545 1465.742

```

This HW model has the (N, N, A) options, i.e., additive error, no trend, and additive seasonality. The optimal value for exponential smoothing constant (alpha) is 0.3601, no smoothing constant for trend estimate (beta = 0), and smoothing constant for seasonality estimate (gamma) is 0.0001.

The model's sales forecast in 12 future months of 2026 is presented below:

	Point	Forecast	Lo 0	Hi 0
Jan 2026		389.5017	389.5017	389.5017
Feb 2026		332.5966	332.5966	332.5966
Mar 2026		310.7888	310.7888	310.7888
Apr 2026		311.8743	311.8743	311.8743
May 2026		306.6674	306.6674	306.6674
Jun 2026		373.0223	373.0223	373.0223
Jul 2026		378.1050	378.1050	378.1050
Aug 2026		389.9754	389.9754	389.9754
Sep 2026		464.7963	464.7963	464.7963
Oct 2026		471.9132	471.9132	471.9132
Nov 2026		488.6605	488.6605	488.6605
Dec 2026		532.7494	532.7494	532.7494

4c. Apply the *accuracy()* function to compare the two models: seasonal naïve forecast (applied in question 3e) and the HW model developed in question 4b. Present the accuracy measures in your report, compare MAPE and RMSE of these forecasts, and identify the best forecasting model.

The accuracy performance measures for the three forecasts are presented below:

Seasonal Naïve Forecast:

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	11.556	48.445	33.769	1.473	12.793	0.598	1.047

HW Model with Automated Model Options and Optimal Parameters:

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	2.187	30.395	21.38	-0.537	7.896	0.003	0.59

The HW model has a substantially better MAPE and RMSE than those for the seasonal naïve forecast. In particular, the margin of forecast error (MAPE) for the HW model, which is 7.90%, is lower than that for the seasonal naïve forecast (12.79%). At the same time, RMSE for the HW model (30.40) is also lower than the same measure for the seasonal naïve forecast (48.45). The conclusion is that the HW model is substantially more accurate than the baseline seasonal naïve forecast.

4d. Compare the best forecasts identified in questions 3c and 4c. Explain what your final choice of the forecasting model in this case will be.

Two-Level Model – Regression and Trailing MA for Residuals:

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	-0.072	20.787	14.751	-0.674	5.258	-0.086	0.404

HW Model with Automated Model Options and Optimal Parameters:

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	2.187	30.395	21.38	-0.537	7.896	0.003	0.59

We compare the MAPE and RMSE measures of the two forecasting models: the two-level model (regression model with linear trend and seasonality and trailing MA for residuals) and HW model with additive error and additive seasonality (A, N, A) and optimal smoothing parameters. The two-level model has lower MAPE (5.26%) and RMSE (20.79) as compared with the HW model's MAPE (7.90%) and RSME (30.40), and thus the two-level model should be the final choice of the forecasting model to project sales in 2026.